Automatic detection of mental arithmetic task and its difficulty level using EEG signals

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***Abstract*—This paper presents a methodology for automatic detection of mental arithmetic tasks and detection of the difficulty level of the given task, using single channel EEG signals. It is vital to understand the response of cerebral cortices to different levels of task difficulty and give a quantitative measure to the role of different brain waves in cognitive tasks. Here, the EEG signals are decomposed using a filter bank to obtain gamma, beta, alpha, theta and delta rhythms and features such as the energy, entropy, mean and L_2 norms are evaluated for each of these rhythms. Performance of three machine learning classifiers namely; support vector machines, decision tree and quadratic discriminant analysis are compared. The algorithm detects the rest state of the brain and the active state of the brain for the given cognitive task. The EEG signals are also classified into two groups based on difficulty level of the given task.**

***Index Terms*—Mental arithmetic task, brain cortices, brain waves, machine learning, EEG**

I. Introduction

Brain is known as an electro-chemical organ and its electrical activity can be recorded as an Electroencephalography (EEG) signal. EEG signals are used in various kinds of applications such as seizure detection as well as prediction, emotion classification, sleep stage classification, and drug effects analysis etc. [1]–[6]. EEG signals can also be used for cognitive task classification [7]–[10], which is the objective of this paper. In order to effectively classify the EEG signals into different categories one must understand the physiology of the brain and the relationship between cognitive task with different cortices of the brain. This can be used in detecting various learning disabilities and in biofeedback systems [11]–[13].

Our cerebral cortex includes frontal, parietal, temporal, and occipital lobes and each of the lobe process distinct sensory information, for example the frontal lobe process the cognitive skills, such as memory, language, judgement, emotional expression, problem solving, and sexual behaviours. The parietal lobes process of sensory information, comprehending spatial orientation, temporal lobe process sound and the occipital lobe process visual information. EEG signals capture the electrical activity of these four cortices and helps in the understanding of function of these cortices [14]. The scalp EEG is commonly acquired using electrodes placed on the scalp. For clinical and research applications, the locations and names of these electrodes are set by the International 10/20 system. In this case, the signals are obtained with 19 electrodes and one ground and system reference electrode [15].

Several authors have studied EEG signals for mental arithmetic tasks to detect the state of brain during rest versus actively involved. Studies have also been conducted to find the functionality of different brain cortices and brain waves for the given task and to study the response of the brain in case of task failure, etc. We are discussing a few such works here: authors in [16] discussed the performance for a mental task activity using causal interaction between θ rhythm of Frontal lobe and α rhythm of parieto-occipital lobe. Yu et al. [17] studied the relation between cortices and the cardiac autonomic nervous activity while a mental arithmetic task is being performed. Wavelet packet parameters and approximate entropy were used as the feature set. In their study, it was observed that the there was an increase in cerebral concious activity with reduction in parasympathetic activity and increase of sympathetic activity. Using EEG signals, Dornhege et al. [18] detected high mental workload in drivers under real traffic conditions when more information was provided.

To detect the rest state or the task state, authors in [19], [20] used statistical features obtained from the wavelet decomposition of EEG signals and machine learning algorithms. Authors in [21] considered alpha rhythm to classify the difficulty level of the given task. Power of alpha wave was computed using Welchs' periodogram. While in [22], alpha and theta waves were used to detect the workload level. Rebsamen et al [23] classified the workload levels using spectral features of the EEG signal. Wang et al., [24] used generalized Higuchi fractal dimension spectrum, power spectral density, autoregressive model and spectral features of EEG signals to recognize mental arithmetic task. The accuracies obtained with single channel is 84.15% and multiple channels is 97.87%.

In [25], frontal EEG signals of twenty healthy subjects were acquired for four cognitive and motor tasks, including arithmetic operation, finger tapping, mental rotation and lexical decision task. Their study concluded that the theta activity increases with difficulty across the four tasks. The accuracy obtained for classifying mental work load level across subjects using support vector machine algorithm was $65\% - 75\%$. In [26], the difficulty level in a mental arithmetic task was detected using alpha rhythm. The dataset considered in their study included the EEG signals from eighteen male subjects.

In this work, we decompose the EEG signals into five

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sub-bands using a set of digital filters such that these sub-band signals capture the natural brain waves namely delta, theta, alpha, beta and gamma. Features like mean, energy and entropy are then extracted from each of these sub-bands and the signal is classified into rest and active state using quadratic discriminant (QD) classifier. From the simulation results obtained in this study, we observe the relationship of each of the brain waves with the cognitive task. This paper also provides a quantitative study of the role of each cortex in the event of a cognitive activity. The paper also present the classification of EEG signal into two groups based on the difficulty of the task. In this case, the EEG signals considered belongs to both active and rest state.

This paper is organized as: Section 2 gives a brief overview of the dataset used in this work, Section 3 discusses the methodology proposed in this work to study the relationship of cognitive task with brain cortices and brain rhythms. In Section 4, we present the simulation results and Section 5 discusses the conclusions of this work and future directions.

II. Dataset

In this work, we have used a publicly available dataset from MIT PhysioNet [27]. The dataset contains EEG recordings from 36 subject after removing the artefacts. The signal has been sampled at 500 Hz and collected using Neurocom monopolar EEG system with twenty three channels. Electrodes were attached at symmetrical anterior frontal, frontal, central, parietal, occipital, and temporal locations corresponding to the international 10/20 scheme. The signals in the given dataset has been pre-processed using a high-pass filter whose cut-off frequency is 0.5Hz, a low-pass filter of 45 Hz passband edge and a power-line interference removal filter. The arithmetic exercise taken upon for this study is the serial subtraction and has been considered as a standardized stress inciting task [28].

The subjects in this study were students of the Taras Shevchenko National University of Kyiv, with their ages in the range of 18 to 26 years. Students without any vision disabilities and with no clinical indication of mental, cognitive, verbal or non-verbal learning impairments were considered. For every participant, the number of operations completed in 4 minutes and accuracy of computation was recorded for performance analysis and for identifying the task difficulty level. The difficulty level was considered low if the result did not vary by more than 20% from the true value. For more detail on the dataset refer [28]. Based on the number of correct arithmetic operations completed per minute, the dataset has been divided into two subsets: group B where the difficulty level is high and group G consisting of participants completing the task without facing any difficulty i.e. reported better computational accuracy. It is pertinent to mention here that in [28] it was mentioned that 12 subjects belong to the group B and 24 subjects to group G, but according to the file "subject-info.csv" given in [27], only 10 subjects were given score 0 and 26 subjects were given score 1. In this work, we have referred to "subject-info.csv" file to get the required groups B and G.

III. Methodology

The rhythmic activity of an EEG signal can be divided into five frequency bands, namely delta, theta, alpha, beta and gamma, where the frequency range of delta rhythm is less than 4Hz, theta rhythm lies between $4-8$Hz, alpha rhythm is between $8-13$Hz, beta lies between $13-30$Hz and gamma rhythm include frequencies above 30 Hz. Each of these brain waves have a particular distribution with respect to cortices and are also linked to different physical activities and emotional states. For instance, high amplitude delta waves are generated during sleep, when the body rejuvenates and rests. Completely alerted brain involved in focussed tasks generates beta waves. The beta wave generation has been associated with concentration and problem solving abilities. Gamma waves are helpful for information preprocessing, learning, perceptions. Alpha waves are helpful to calm down and promotes to feel more relax. Theta waves are related to the emotions, intuition, and creative thinking.

We obtain the EEG rhythms termed as gamma, beta, alpha, theta and delta using a bank of digital filters. These brain rhythms are the sub-band signals obtained via the frequency decomposition of the EEG signals as shown in Fig. 1, where $H_i(z)$ denotes the digital filters.

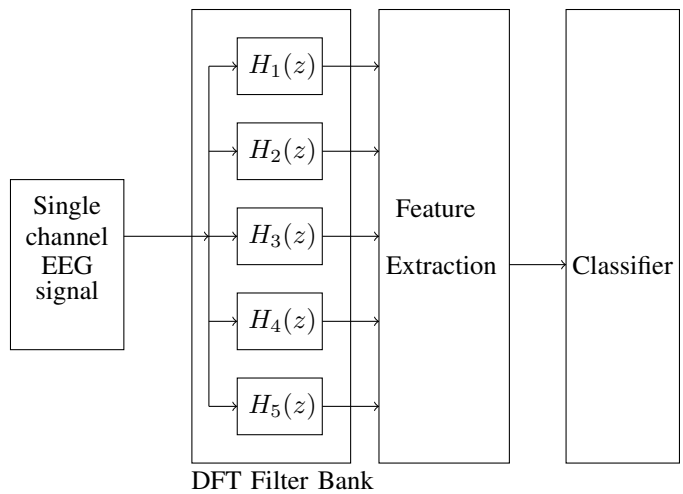

Fig. 1. Block Diagram of the proposed methodology

The output of the digital filters, as shown in Fig. 1, are the five sub-band signals or brain rhythms denoted by $x_i[n]$ for $0 \leq i \leq 4$. We obtain features like energy, entropy, mean and L_2 norm, [29] for each of these signal as given below:

$$Entropy = -\sum_{n=0}^{N-1} p(x_i[n]) \log_2(p(x_i[n])), \tag{1}$$

where $p(x_i[n])$ is the discrete probability of $x_i[n]$,

$$Energy = \sum_{n=0}^{N-1} |x_i[n]|^2 \tag{2}$$

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$$L_2 norm = \left(\sum_{0}^{N-1} |x_i[n]|^2 \right)^{1/2} \quad (3)$$

and

$$Mean = \frac{1}{N} \sum_{n=0}^{N-1} (x_i[n]), \quad (4)$$

These extracted features are then given to machine learning classifiers namely; SVM, decision tree and quadratic discriminant. A detail discussion on these learning algorithms can be found in [30]. In the next section, we present the simulation results obtained using the proposed algorithm.

IV. Simulations

In this section, we present two results obtained using the proposed methodology. The first experiment is designed to study the role of each cortex in cognitive tasks. Here, we present the results for the classification of EEG signals recorded from different cortices into active and rest classes. The second experiment classifies the EEG signals based on the difficulty level of the task. The EEG signals from both active and rest states are classified and the signals from all the channels are considered in this experiment. The results obtained from both experiments are given below:

Result 1: Here, our aim is to classify the EEG signals from the dataset into rest and active state. Fig. 2- Fig.3 shows the power spectral density of a sample EEG signal belonging to the set of rest state and an active state of group G and B. the power spectral. In Table I, we present the classification accuracies of three classifiers namely; SVM, decision tree, and QD when the feature set includes entropy, energy and mean of data from all the sub-bands. In Table II, we present the classification results for each sub-band for different cortices. In the name of the channels, F refers to Frontal lobe, T refers to Temporal lobe, P refers to Parietal lobe and O is for Occipital lobe.

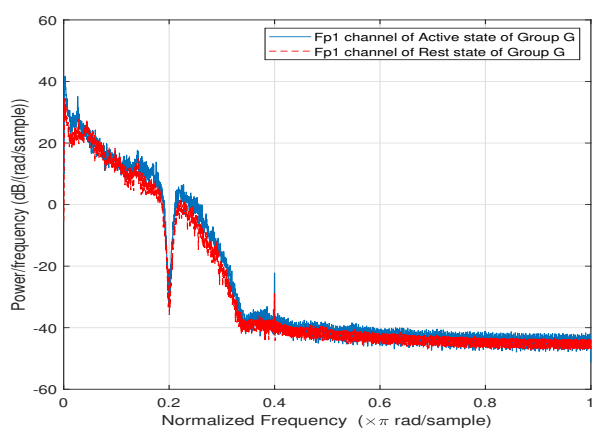

Fig. 2. Power spectral density of EEG signals from group G for rest and active state

Result 2: Here, our aim is to detect the brain response to the task difficulty of the cognitive activity based on the EEG signals acquired both before the task and after the task. Fig. 4 shows the power spectral density of a sample EEG signal belonging to the set of group G and B.There are two difficulty levels labelled in the dataset as group B and G. EEG signal recorded from each channel is decomposed into the five brain waves, as discussed in section III. From each rhythm, features like energy, entropy, mean and L_2 norm are computed. The feature set is then used to classify each EEG signal using quadratic discriminant classifier, the results are tabulated in Table III. Here, group B is comprised of only 10 subjects and therefore, to balance the dataset we have used re-sampling with replacement operation to increase the dimensionality of this group to 26.

Table IV provides classification results obtained from the literature for detection of cognitive tasks using EEG signals, however, It should be noted that the works mentioned in Table

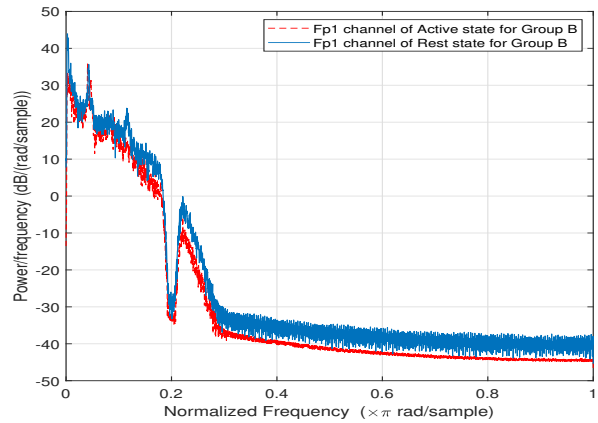

Fig. 3. Power spectral density of EEG signals from group B for rest and active state

TABLE I
Classification accuracies for detecting mental task using EEG signals

Channel	Accuracy (%)		
	SVM	Decision Tree	Quadratic Discriminant
Fp1	80.6	90.3	94.4
Fp2	83.3	81.9	97.2
F3	81.9	90.3	100
F4	66.7	90.3	94.4
F7	81.9	84.7	94.4
F8	75	87.5	98.6
T3	77.8	91.7	97.2
T4	66.7	90.3	95.8
C3	80.6	83.3	97.2
C4	79.2	88.9	95.8
T5	72.2	88.9	98.6
T6	76.4	90.3	93.1
P3	75	94.4	97.2
P4	75	86.1	97.2
O1	75	95.8	95.8
O2	65.3	84.7	95.8
Fz	80.6	87.5	100
Cz	81.9	93.1	97.2
Pz	77.8	88.9	97.2

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TABLE II
CLASSIFICATION ACCURACIES OF DETECTING MENTAL TASK USING FEATURES OF BRAIN RHYTHMS

	Accuracy (%)			
Channel	sub-band	SVM	Decision Tree	Quadratic Discriminant
Fp1	delta	69.4	66.7	75
Fp1	theta	69.4	70.8	84.7
Fp1	alpha	91.7	91.7	91.7
Fp1	beta	80.6	83.3	88.9
Fp1	gamma	77.8	70.8	76.4
Fp2	delta	70.8	81.9	76.4
Fp2	theta	70.8	84.7	80.6
Fp2	alpha	61.1	87.5	81.9
Fp2	beta	76.4	79.2	87.5
Fp2	gamma	73.6	80.6	73.6
F4	delta	66.7	79.2	72.2
F4	theta	79.2	80.6	84.7
F4	alpha	79.2	90.3	90.3
F4	beta	80.6	97.2	93.1
F4	gamma	77.8	72.2	79.2
F3	delta	70.8	73.6	73.6
F3	theta	72.2	83.3	80.6
F3	alpha	80.6	83.3	88.9
F3	beta	88.9	93.1	95.8
F3	gamma	77.8	68.1	73.6
F7	delta	65.3	65.3	66.7
F7	theta	70.8	88.9	88.9
F7	alpha	83.3	87.5	88.9
F7	beta	81.9	90.3	91.7
F7	gamma	70.8	88.9	70.8
F8	delta	66.7	65.3	70.8
F8	theta	77.8	79.2	79.2
F8	alpha	75	83.3	83.3
F8	beta	84.7	86.1	86.1
F8	gamma	77.8	79.2	80.6
T3	delta	63.9	58.3	69.4
T3	theta	80.6	83.3	84.7
T3	alpha	76.4	86.1	88.9
T3	beta	87.5	93.1	94.4
T3	gamma	72.2	84.7	83.3
Cz	delta	63.9	66.7	70.8
Cz	theta	86.1	90.3	81.9
Cz	alpha	88.9	97.2	93.1
Cz	beta	88.9	95.8	97.2
Cz	gamma	86.1	90.3	87.5
P4	delta	73.6	68.1	72.2
P4	theta	72.2	79.2	75
P4	alpha	81.9	91.7	94.4
P4	beta	79.2	86.1	88.9
P4	gamma	73.6	81‘.9	80.6
O2	delta	69.4	77.8	70.8
O2	theta	75	84.7	79.2
O2	alpha	80.6	84.7	86.1
O2	beta	80.6	88.9	90.3
O2	gamma	73.6	77.8	73.6
O1	delta	77.8	76.4	72.2
O1	theta	72.2	79.2	88.9
O1	alpha	83.3	98.6	97.2
O1	beta	79.2	90.3	90.3
O1	gamma	69.4	72.2	77.8

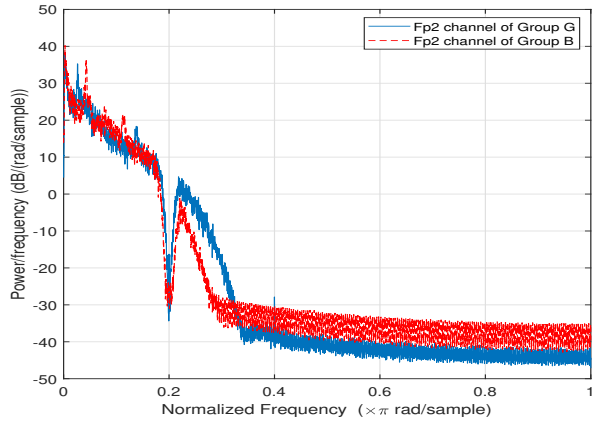

Fig. 4. Power spectral density of EEG signals from group G for rest and active state

TABLE III
CLASSIFICATION ACCURACY OF OBTAINED FOR DETECTING GROUP B AND G

Channel No.	Rest State	Active State
Fp1	82.7	84.6
Fp2	88.5	90.4
F3	76.9	78.8
F4	80.8	80.8
F7	76.9	88.5
F8	84.6	82.7
T3	90.4	90.4
T4	88.5	94.2
C3	88.5	92.3
C4	92.3	92.3
T5	82.7	88.5
T6	84.6	80.8
P3	82.7	94.2
P4	90.4	88.5
O1	92.3	84.6
O2	86.5	82.7
Fz	84.6	88.5
Cz	94.2	88.5
Pz	90.4	90.4

TABLE IV
COMPARISON OF PROPOSED METHODOLOGY AND STATE-OF-ART ALGORITHMS FOR DETECTING MENTAL ARITHMETIC TASK USING EEG SIGNALS

S.No.	Authors	number of tasks	No. of channels	Accuracy
1	Ramaswamy [31]	2	6	94.4%
2	Del R Millan et.al. [32]	3	8	70%
3	Wang et.al. [24]	2	1	84.15%
4	Anderson et.al. [33]	2	6	91.4%
5	Liang N et al. [34]	5	6	51%

IV have each considered different data-sets.

V. CONCLUSIONS

In this paper, we have presented a study on the relationship of mental arithmetic task and brain cortices. We classified the given EEG signal into rest state and active state using features like entropy, energy and mean of the EEG signal. In this work, we decomposed the data into five sub-bands using a set of digital filters such that the output of these filters provides the brain rhythms delta, theta, alpha, beta and gamma. This paper explores the role of each brain rhythm in detecting the

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rest state and the active state for the given cognitive task. We also proposed an algorithm to detect the level of difficulty of the given task using EEG signals. This work can be used for brain computer interface algorithms and also in providing an insight into the brain responses for various learning disorders. As future work we will test the robustness of the proposed algorithm on more raw EEG datasets collected at different sampling rates.

REFERENCES

[1] Z. Lasefr, S. S. V. N. R. Ayyalasomayajula, and K. Elleithy, "Epilepsy seizure detection using EEG signals," *2017 IEEE 8th Annual Ubiquitous Computing, Electronics and Mobile Communication Conference (UEMCON), New York, NY*, pp. 162–167, Oct 2017.

[2] Y. Park, T. Netoff, and K. Parhi, "Seizure prediction with spectral power of time/space-differential EEG signals using cost-sensitive support vector machine," *2010 IEEE International Conference on Acoustics, Speech and Signal Processing, Dallas, Texas*, pp. 5450–5453, Mar 2010.

[3] J. Preethi and S. Sowmiya, "Emotion recognition from EEG signal using ISO-FLANN with firefly algorithm," *2016 International Conference on Communication and Signal Processing (ICCSP), Melmaruvathur*, pp. 1932–1936, Apr 2016.

[4] K. Aboalayon, M. Faezipour, W. Almuhammadi, and S. Moslehpour, "Sleep stage classification using EEG signal analysis: A comprehensive survey and new investigation," *Entropy*, vol. 18(9), p. 272, Aug 2016.

[5] W. Banoczi, "How some drugs affect the electroencephalogram EEG," *American journal of electroneurodiagnostic technology*, vol. 45, pp. 118–29, Aug 2005.

[6] I. Seleznov, I. Zyma, K. Kiyono, S. Tukaev, A. Popov, M. Chernykh, and O. Shpenkov, "Detrended fluctuation, coherence, and spectral power analysis of activation rearrangement in eeg dynamics during cognitive workload," *Frontiers in Human Neuroscience*, vol. 13, p. 270, 2019.

[7] N. Friedman, T. Fekete, K. Gal, and O. Shriki, "EEG-based prediction of cognitive load in intelligence tests," *Frontiers in Human Neuroscience*, vol. 13, p. 191, 2019. [Online]. Available: https://www.frontiersin.org/article/10.3389/fnhum.2019.00191

[8] M. A. Almogbel, A. H. Dang, and W. Kameyama, "Cognitive workload detection from raw EEG-signals of vehicle driver using deep learning," *2019 21st International Conference on Advanced Communication Technology (ICACT)*, pp. 1–6, 2019.

[9] T. Fernndez, T. Harmony, M. Rodrguez, J. Bernal, J. Silva, A. Reyes, and E. Marosi, "EEG activation patterns during the performance of tasks involving different components of mental calculation," *Electroencephalography and Clinical Neurophysiology*, vol. 94, no. 3, pp. 175 – 182, 1995. [Online]. Available: http://www.sciencedirect.com/science/article/pii/001346949400262J

[10] J. K. Nuamah, Y. Seong, and S. Yi, "Electroencephalography (EEG) classification of cognitive tasks based on task engagement index," *2017 IEEE Conference on Cognitive and Computational Aspects of Situation Management (CogSIMA)*, pp. 1–6, 2017.

[11] K. K. Ang, C. Guan, K. Lee, J. Q. Lee, S. Nioka, and B. Chance, "A brain-computer interface for mental arithmetic task from single-trial near-infrared spectroscopy brain signals," *2010 20th International Conference on Pattern Recognition*, pp. 3764–3767, 2010.

[12] S. Dimitriadis, Y. Sun, N. Thakor, and A. Bezerianos, "Causal interactions between frontal parieto-occipital2 predict performance on a mental arithmetic task," *Frontiers in Human Neuroscience*, vol. 10, Sep 2016.

[13] R. Rothschild, "Neuroengineering tools/applications for bidirectional interfaces, braincomputer interfaces, and neuroprosthetic implants a review of recent progress," *Frontiers in neuroengineering*, vol. 3, p. 112, Oct 2010.

[14] B. T. Yeo, F. Krienen, J. Sepulcre, M. Sabuncu, D. Lashkari, M. Hollinshead, J. Roffman, J. Smoller, L. Zollei, J. Polimeni, B. Fischl, H. Liu, and R. Buckner, "The organization of the human cerebral cortex estimated by functional correlation," *Journal of neurophysiology*, vol. 106, pp. 1125–65, Jun 2011.

[15] R. W. Homan, "The 10-20 electrode system and cerebral location," *American Journal of EEG Technology*, vol. 28, no. 4, pp. 269–279, 1988.

[16] S. I. Dimitriadis, Y. Sun, N. V. Thakor, and A. Bezerianos, "Causal interactions between frontal $\theta-$parieto-occipital $\alpha 2$ predict performance on a mental arithmetic task," *Frontiers in Human Neuroscience*, vol. 10, p. 454, 2016. [Online]. Available: https://www.frontiersin.org/article/10.3389/fnhum.2016.00454

[17] X. Yu, J. Zhang, D. Xie, J. Wang, and C. Zhang, "Relationship between scalp potential and autonomic nervous activity during a mental arithmetic task," *Autonomic neuroscience : basic & clinical*, vol. 146, pp. 81–6, Feb 2009.

[18] G. Dornhege, J. del R. Milln, T. Hinterberger, D. J. McFarland, and K. Mller, *Improving Human Performance in a Real Operating Environment through Real-Time Mental Workload Detection*. MITP, 2007, pp. 409–422. [Online]. Available: https://ieeexplore.ieee.org/document/6281209

[19] H. U. Amin, A. Malik, R. F. Ahmad, N. Badruddin, N. Kamel, M. Hussain, and W.-T. Chooi, "Feature extraction and classification for eeg signals using wavelet transform and machine learning techniques," *Australasian Physical and Engineering Sciences in Medicine*, vol. 38, Feb 2015.

[20] N. S. Pathan, M. Foysal, and M. M. Alam, "Efficient mental arithmetic task classification using wavelet domain statistical features and svm classifier," *2019 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, pp. 1–5, Feb 2019.

[21] E. Magosso, F. De Crescenzio, G. Ricci, S. Piastra, and M. Ursino, "Eeg alpha power is modulated by attentional changes during cognitive tasks and virtual reality immersion," *Computational Intelligence and Neuroscience*, vol. 2019, pp. 1–18, Jun 2019.

[22] A. Gevins and M. E. Smith, "Neurophysiological measures of cognitive workload during human-computer interaction," *Theoretical Issues in Ergonomics Science*, vol. 4, no. 1-2, pp. 113–131, 2003. [Online]. Available: https://doi.org/10.1080/14639220210159717

[23] B. Rebsamen, K. Kwok, and T. Penney, "Evaluation of cognitive workload from eeg during a mental arithmetic task," *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, vol. 55, pp. 1342–1345, Sep 2011.

[24] Q. Wang and O. Sourina, "Real-time mental arithmetic task recognition from eeg signals," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 21, no. 2, pp. 225–232, Mar 2013.

[25] W. So, S. Wong, J. Mak, and R. Chan, "An evaluation of mental workload with frontal eeg," *PLOS ONE*, vol. 12, p. e0174949, Apr 2017.

[26] F. Al-Shargie, *Mental Stress Quantification Using EEG Signals*, Jan 2016, pp. 29–33.

[27] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.

[28] I. Zyma, S. Tukaiev, I. Seleznov, K. Kiyono, A. Popov, M. Chernykh, and O. Shpenkov, "Electroencephalograms during mental arithmetic task performance," *Data*, vol. 4, Jan 2019.

[29] B. Fatimah, P. Singh, A. Singhal, and R. B. Pachori, "Detection of apnea events from ecg segments using fourier decomposition method," *Biomedical Signal Processing and Control*, vol. 61, p. 102005, 2020. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S1746809420301610

[30] G. Bonaccorso, *Machine Learning Algorithms: A Reference Guide to Popular Algorithms for Data Science and Machine Learning*. Packt Publishing, 2017.

[31] R. Palaniappan, "Utilizing gamma band to improve mental task based brain-computer interface design," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 14, no. 3, pp. 299–303, Sep 2006.

[32] J. del R. Millan, J. Mourino, F. Babiloni, F. Cincotti, M. Varsta, and J. Heikkonen, "Local neural classifier for eeg-based recognition of mental tasks," *Proceedings of the IEEE-INNS-ENNS International Joint Conference on Neural Networks. IJCNN 2000. Neural Computing: New Challenges and Perspectives for the New Millennium*, vol. 3, pp. 632–636 vol.3, Jul 2000.

[33] C. W. Anderson, E. A. Stolz, and S. Shamsunder, "Multivariate autoregressive models for classification of spontaneous electroencephalographic signals during mental tasks," *IEEE Transactions on Biomedical Engineering*, vol. 45, no. 3, pp. 277–286, Mar 1998.

[34] N.-Y. Liang, P. Saratchandran, G.-B. Huang, and N. Sundararajan, "Classification of mental tasks from eeg signals using extreme learning

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machine," *International journal of neural systems*, vol. 16 1, pp. 29–38, 2006.

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